Environmental Sensor Data Visualization Using PCA & UMAP

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INTRODUCTION

- Environmental monitoring generates a large amount of air-quality and meteorological sensor data. Analyzing multiple sensor measurements together can be difficult because the data contains many features. In this project, 13 numerical environmental features including CO, NOx, NO_2-related sensor measurements, temperature, relative humidity and absolute humidity are analyzed. PCA and UMAP are used to reduce the high-dimensional data into two dimensions for easier visualization and comparison. K-Means clustering is then applied to the UMAP representation to identify groups of similar observations.

ABSTRACT

- Environmental sensor data contains multiple measurements that are difficult to analyze and visualize together. This project uses PCA and UMAP to reduce 13 environmental features into two-dimensional representations. The data is cleaned, missing values are handled, and the features are standardized before dimensionality reduction. PCA and UMAP visualizations are generated and compared to observe how the data is represented in two dimensions. K-Means clustering is then applied to the UMAP representation to form four groups. Finally, the original environmental features are analyzed cluster-wise by calculating average values and comparing them with the overall average to describe the characteristics of each cluster.

EXISTING SYSTEM

Traditional environmental data analysis mainly uses tables, individual graphs, and comparisons between sensor measurements. When many environmental features are analyzed together, it becomes difficult to visualize relationships and identify groups in high-dimensional data. PCA and UMAP can simplify this data by reducing its dimensions. However, the dimensions of a UMAP representation are not directly connected to the original environmental variables, making interpretation of the resulting groups less direct. Therefore, additional analysis of the original feature values is useful for understanding the characteristics of each group.

PROPOSED SYSTEM

The proposed system loads the Air Quality UCI dataset and selects 13 numerical environmental features. The data is cleaned by removing unnecessary empty columns, replacing -200 values with missing values, and filling missing values using the median. The features are then standardized before applying PCA and UMAP for two-dimensional visualization. K-Means clustering is applied to the UMAP representation to create four groups. Finally, the original environmental features are analyzed for each cluster and compared with the overall average to describe the characteristics of the groups.

FLOW DIAGRAM

- Dataset
- ↓
- Preprocessing
- ↓
- Feature Selection
- ↓
- Standardization
- ↓
- PCA + UMAP
- ↓
- Visualization & Comparison
- ↓
- K-Means Clustering
- ↓
- Cluster Analysis

ALGORITHM

- Load the Air Quality UCI dataset.
- Preprocess the data and handle missing values.
- Select 13 numerical environmental features.
- Standardize the selected features.
- Apply PCA and UMAP to reduce the data to 2 dimensions.
- Generate and compare the PCA and UMAP visualizations.
- Apply K-Means clustering on the UMAP representation.
- Form 4 clusters and analyze the original feature values for each cluster.
- Compare cluster averages with the overall averages to describe cluster characteristics.

EXPECTED OUTPUT

- The system is expected to generate two-dimensional PCA and UMAP visualizations of the environmental sensor data. The PCA and UMAP plots can be compared to observe how the observations are distributed and whether similar observations appear close together. K-Means clustering on the UMAP representation produces four groups. The original environmental features are then analyzed for each group to identify relatively higher and lower feature values and describe the characteristics of the clusters.

THANKYOU